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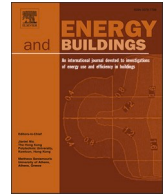


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Unraveling energy justice in NYC urban buildings through social media sentiment analysis and transformer deep learning

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ABSTRACT

This study explores the intricate relationship between human sentiment on social media data, herein tweet posts on X platform, urban building characteristics, and the socio-spatial dynamics of New York City (NYC) boroughs. Leveraging Natural Language Processing (NLP) techniques, particularly sentiment analysis, augmented by the capabilities of transformer deep learning models, RoBERTa, the study places particular emphasis on the term ‘Stay-at-Home’ to encapsulate the pronounced shift in building occupancy during the pandemic’s inaugural year. This focus intertwines with pivotal terms like ‘Energy Bill’ and ‘HVAC’, shedding light on their interconnected implications. The sentiment analysis leverages data from New York City’s PLUTO and the Department of Energy’s LEAD databases to emotional disparities connected to urban building characteristics as well as demographic and socioeconomic factors. This analytical approach unravels prevailing public emotions and extends the discussion to include energy justice concerns, viewing them through the lens of the city’s built infrastructure. The research uncovers profound disparities in the built environment and the allocation of resources in NYC, highlighting the critical need to embrace a spatial justice framework for a sustainable future. This research can aid designers, planners, and policymakers in their efforts to promote equitable and inclusive urban development.

1. Introduction

Social media platforms provide a dynamic data source for capturing perceptions and subjective views within built environments, actively influencing how individuals perceive their lives and the surrounding world [1]. Such platforms promote participation, interaction, and open dialogue, enabling a myriad of stakeholders to present varying narratives [2]. Grounded in Marshal McLuhan’s assertion that media technologies are extensions of human cognition, affecting our interactions with our surroundings [3,4], social media stands as a testament to this belief, offering platforms for expansive public discourse. These platforms’ real-time insights into popular opinion make them indispensable research tools [5]. The immense data found on social media provides a primary avenue for exploring individuals’ emotions, sentiments, and experiences [6]. Recently, using social media to monitor and assess public perception has emerged as a vital method in environmental research. This influence extends to environmental activism, where social

networks fill the gaps left by the absence of formal legal systems, potentially shaping ecosystem governance [7]. Such feedback can determine the trajectory towards sustainability and the success of interventions [8]. Notably, studies have delved into platforms like X framing significant environmental events, highlighting the indispensable role of digital mediums in crafting environmental narratives [9].

Addressing data scarcity in environmental studies, social media emerges as an unconventional, yet insightful reservoir of data [10–14]. Users’ spontaneous posts on platforms, notably X, inadvertently offer real-time public insights, spotlighting environmental issues often missed by traditional measures [15]. Social media’s capacity to act as an “on-the-ground monitor” [16] and a “social sensor” [17] underscores its pivotal role in melding data capture with lived experiences. These characteristics creates a two-way impact – social media acts as both a source of information and a catalyst for change, driving public engagement and government intervention to mitigate health risks associated with air pollution.

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Amidst the rise of stay-at-home orders, social media data has emerged as a pivotal instrument for scrutinizing feedback on a diverse array of topics related to the built environment. This includes aspects such as indoor air quality [18] and shifts in behavioral patterns on social platforms [19], among others. Social media's key role in shaping urban policies and promoting spatial justice, especially in the context of energy, cannot be ignored [20]. It serves as a platform for both detecting disparities and advocating for change, making it a critical tool in driving energy equity in urban contexts. Nevertheless, while the significance of social media data in comprehending energy behavior within urban spaces is well-documented [21–23], these studies predominantly delve into the temporal and spatial dimensions of social media engagements, with a pronounced focus on mobility patterns. This research approach often overlooks an exploration of the actual content of posts. In essence, there's a prevalent oversight in leveraging text mining techniques to extract insights about energy-related concerns from social media discourse during stay-at-home period.

Energy justice embodies the pursuit of equitable access to energy resources, ensuring that all individuals have access to affordable, reliable, and sustainable energy. It builds on the principles of environmental justice, which seeks fair treatment and meaningful involvement of all people regardless of race, color, national origin, or income with respect to environmental laws, regulations, and policies. The concept was more clearly defined and articulated through the work of Schlosberg [24], who viewed 'justice' in environmental scenarios as inclusive of equitable distribution, recognition, and procedural participation. This notion has since fostered an interdisciplinary dialogue, drawing attention to the importance of integrating energy justice into educational curricula and policy discussions, as noted by [25] and [26]. This integration is seen as essential to ensuring that future policymakers and economists are equipped to craft energy policies that are fair and consider the broader environmental and climate context.

The empirical dimension of energy justice has been explored through studies that analyze the intersection of energy access with socioeconomic and geographic factors. For instance, research by [27] spotlights the spatial aspects of energy poverty, indicating that location can significantly influence an individual's energy burden. Socioeconomic analyses, such as those by [28], have uncovered correlations between race, ethnicity, and energy efficiency, with particular emphasis on how these factors contribute to fuel poverty in urban neighborhoods. Broadening the scope, a regional study by [29] quantifies the extent of energy injustice, reporting that a substantial segment of the population is financially strained by high energy costs. These studies collectively point to the necessity for comprehensive policy measures that address energy justice, ensuring that energy policies do not inadvertently perpetuate or exacerbate social inequities.

Most recently, study by [30] report on the prevalent issue of energy inequity in the United States, emphasizing that a considerable segment of households grapple with energy poverty, notably within minorities. The study underscores the necessity to augment support measures and advocates for a more encompassing federal categorization geared towards energy poverty assistance. In an integrated approach, study by [31] introduce a framework called iEnEx, which amalgamate household energy burden with the spatiotemporal dynamics of air pollution exposure. This framework, tested using a big data approach with land use regression and machine learning for Chicago, demonstrates its utility in directing stakeholders towards inclusive and equitable urban development. Nonetheless, the exploration of social media data to understand public perception of energy burden, particularly during periods of increased home occupancy during the pandemic, remains underexplored.

The evolution of data-driven approaches has led to enhanced models that center human considerations in planning built environments. Among these, text mining stands out by employing Natural Language Processing (NLP) to interpret human language, transforming unstructured text into a format ready for analysis [32]. NLP has its roots in the

mid-20th century, born from the intersection of linguistics and artificial intelligence [33–36]. It was initially distinct from the field of text information retrieval (IR), which uses statistical methods for indexing and searching large databases of text [37,38]. While IR focuses on text searchability, NLP's broader scope includes the contextual interpretation of language. Notably, text mining, a practical application of NLP, has proved crucial for the efficient collation of relevant information in a cost-effective manner [39].

In computational analyses of social media using NLP techniques, a dominant reliance on traditional rule-based methodologies, particularly lexicon-based sentiment analysis, is evident when deciphering human sentiments on the built environment concerns [40]. Specifically, linguistic models in energy-focused research seem limited, as most studies predominantly employ lexicon-driven methods, often based on extensive journal publication databases [41]. Yet, there's a distinct lack of research that integrates machine learning (ML) techniques while utilizing textual data from social media platforms to capture real-world sentiments about urban energy systems including urban building operational systems and energy inequities in cities. This highlights a significant gap, especially when considering the depth and authenticity of insights that deep learning can offer, aligning more closely with real-world data. The potential of modern pre-trained models, especially those developed from comprehensive real-world social media datasets like the X platform using transformer-based deep learning [42], remains largely untapped in the area.

This study aims to delve deeper into social media content, moving beyond the standard spatial and temporal analyses, to extract meaningful insights from users' posts. Despite a growing body of research on energy burdens and urban disparities, there's a noticeable lack of studies using social media to gauge public sentiment, especially during significant events like stay-at-home mandates. Leveraging data-driven techniques and the capabilities of NLP, particularly sentiment analysis approaches, this research intends to shift from the traditional lexicon-based methods to more emerging deep learning techniques. Specifically, we plan to employ the RoBERTa [43] transformer-based deep learning model on comprehensive social media data from New York City boroughs. This will provide a deeper understanding of urban building energy drivers, such as HVAC systems, and the associated energy burdens. Ultimately, our aim is to address the aforementioned research gaps, ensuring that our analysis of social media accurately reflects public opinion and supports informed policy decisions promoting energy equity in urban areas.

2. Materials and methods

This research entails three main steps, outlined in Fig. 1. The initial step involves retrieving data from various sources, with a detailed explanation of this process provided in the subsequent sections. The second stage includes NLP analysis, where the collected data undergoes different processes and analyses. The final stage of exploration investigates the relationships between the sentiment analysis outcomes of the individual terms and the physical and semantic characteristics of urban buildings, as well as the societal dynamics within different city districts, providing a holistic understanding of the interplay between public sentiment and urban environmental and demographic factors. It should be noted that this exploration handles relationships between each term and other urban aspects separately, as conducting a comprehensive analysis to statistically explore the interrelations between the 'Stay-at-Home', 'Energy Bill', and 'HVAC' datasets typically necessitates a multi-dimensional analysis which is optimally utilized on a single dataset where each data point comprises a full set of categorical responses. Notably, each term's relationship with urban elements is analyzed independently to preserve the integrity and precision of the findings and to prevent any misconceptions that might emerge from combining datasets with disparate sizes and non-aligned data points.

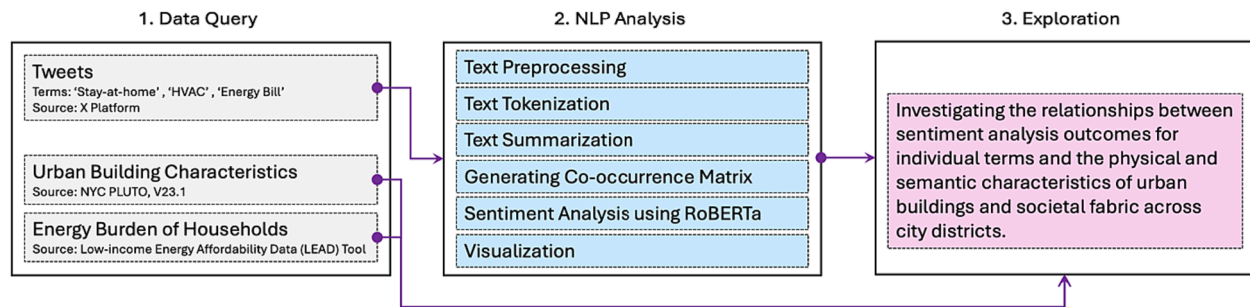


Fig. 1. Flowchart illustrating the three-phase methodology of the study, starting with data query, followed by NLP analysis, and concluding with the exploration of relationships between sentiment analysis results, urban building characteristics, and the societal context across city districts.

2.1. Data query

2.1.1. Retrieving data from the X platform

In this study, we selected X (formerly Twitter) as the source for social media data for several reasons. X users send short messages limited to 280 characters, allowing sentiments to be conveyed succinctly. Additionally, X offers an API key to access its entire historical data. Many studies have also chosen X as their primary social media platform for human-centered explorations [44], as highlighted by a recent comprehensive review by [11]. We utilized the *academictwitteR* package in R [45], combined with our X API v2 key, to extract historical data. In our georeferenced data acquisition process, we identified three primary keywords: 'stay-at-home', 'energy bill', and 'HVAC'. Our aim was to investigate the interplay between sentiments related to energy burdens and the sentiments about urban building operational systems across NYC boroughs, especially in the context of concerns surrounding the 'stay-at-home' orders. For the term 'stay-at-home', we sourced data spanning the initial year of the stay-at-home order in NYC, which ran from March 1st, 2020, to February 28th, 2021.

"In contrast, the terms 'energy bill' and 'HVAC' warranted a broader timeframe, given their relevance beyond the context of COVID's stay-at-home orders. As such, we gathered data for these topics over an 8-year period, starting from January 1st, 2015, and extending through December 2022. Our selection of the terms 'HVAC', 'Energy Bill', and 'Stay-at-Home' as focal points in our research was based on their direct relevance to indoor living conditions, particularly during the pandemic. 'HVAC' systems ensure maintaining indoor environmental quality (IEQ), particularly indoor air quality (IAQ) and thermal comfort. 'Energy Bills' reflect the financial cost of sustaining these indoor conditions. The 'Stay-at-Home' orders have markedly influenced both the usage and perception of these aspects amidst the lockdown. Our objective was to provide a comprehensive understanding of public sentiment regarding indoor living conditions, accounting for potential biases and platform dynamics. However, we acknowledge that 'HVAC' might be more familiar to individuals with architectural or engineering backgrounds, potentially introducing bias.

These terms were queried based on exact phrases, which identified tweets that included only these terms within the posted comments to highlight the subject matter. To avoid bots, we only implemented data acquisition based only on verified X accounts. To determine X users' genders, which aren't listed on their profiles, we used a gender classification tool that analyzes usernames. This tool takes the first name-like string from the username, after removing common words, and compares it to a database of names given to babies in the United States, as recorded by the Social Security Administration. We used 'gender' package [46] in R to perform the task. This method has been applied by [47] for gender analysis of X data for indoor air quality (IEQ) in office buildings in the U. S."

To ascertain the geographical context of tweets, we employed the 'place' parameter available in the *academictwitteR* package. This parameter is integral for filtering tweets based on location information

that is either explicitly geotagged by users or implicitly associated with their tweets. Utilizing the 'place' parameter allowed us to refine our data collection process to include only those tweets that were tagged with geographical identifiers corresponding to specific locations, such as cities or neighborhoods. This approach enhances the precision of our geographical analysis by leveraging Twitter's geolocation tagging feature, which tends to be more reliable than self-reported profile locations. It enables the aggregation of data that reflects the actual location from which tweets are posted, thereby providing a robust dataset for analyzing spatial patterns in Twitter activity. By implementing this parameter in our data retrieval, we minimized the potential for error due to inaccurate self-reporting of locations and ensured that the geolocation information in our dataset was consistent with the users' actual tweeting locations.

2.1.2. N.Y.C. Urban building characteristics

In this research, we used the NYC PLUTO (Primary Land Use Tax Lot Output) dataset, which is a comprehensive collection of land use and geographic data pertaining to the five boroughs of New York City: Manhattan, Brooklyn, Queens, The Bronx, and Staten Island. It is maintained and released by the NYC Department of City Planning. PLUTO provides detailed information on tax lots, which are individual pieces of land that are used as the basis for property taxes. The dataset includes a wealth of attributes for each lot, such as its land use category, building type, address, owner, zoning information, building geometries, year of construction, and other characteristics. One of the unique features of the PLUTO dataset is its granularity. It doesn't just provide city's zoning and land use information; it depicts specifics about individual properties, making it a valuable resource exploring the built environment of New York City. Beyond its primary functions, PLUTO has been used for various analytical purposes, including urban building energy modeling (UBEM) frameworks [48,49]. Its regular updates ensure that users have access to the most current snapshot of New York City's ever-evolving landscape. In this research, we utilized version 23v2 of the PLUTO database.

2.1.3. Energy burden of households

We utilized the Low-income Energy Affordability Data (LEAD) Tool [50], provided by the U.S. Department of Energy's Office of Energy Efficiency & Renewable Energy (EERE, 2020). This dataset offers a granular look at the energy burden on low-income households across the United States, quantifying the proportion of income that these households allocate to energy expenses. The LEAD Tool aggregates data down to the census tract level, enabling a detailed examination of energy affordability that is pivotal to our analysis. The Tool further provides detailed data on building ages, highlighting the energy efficiency variations between older and newer structures. It also catalogues diverse energy sources, from traditional fuels such as coal to renewables such as solar and wood. Crucially, it introduces an "energy burden" metric, ranging from 1 to 15, representing the percentage of household income spent on energy; a lower value indicates lesser expenditure on energy

relative to income. The tool categorizes income groups using standard income level classifications typically defined as Area Median Income (AMI), State Median Income (SMI), and Federal Poverty Level (FPL). For the purposes of our study, we extracted and then aggregated data for the AMI category to the borough (county) level within New York City, offering a nuanced perspective on energy affordability crucial for the analysis. This dataset is well-established in scholarly research for examining energy inequity on an urban scale [31].

2.2. NLP analytics

In the methodology section, our focus is on NLP analytics, which is pivotal in comprehending and interpreting vast amounts of textual data. In line with standard NLP practices, we initiate our process with text preprocessing, a step where raw data is cleaned and standardized to enhance the reliability of analysis. This is succeeded by text tokenization, segmenting text into smaller, manageable units. We subsequently engage in text summarization, a technique aimed at condensing extensive texts into their core summaries. An integral part of our approach involves generating co-occurrence matrices, instrumental in revealing word relationships within our dataset. For sentiment analysis, we utilize a Transformer-based deep learning model, RoBERTa, renowned for its adeptness in grasping context and language subtleties, particularly in social media content. Our methodology is further enriched by advanced visualization techniques: embedding visualization for deciphering word relationships in a multi-dimensional framework, and attention score visualization for insights into our model's focal points during text analysis. These methodologies synergize to form the foundation of our NLP analytics, enabling the extraction of patterns and sentiments from textual data.

2.2.1. Text preprocessing

We employed a preprocessing pipeline on the text data before introducing it to the model, ensuring its consistency and eliminating any noise. Table 1 enumerates the twelve methods we utilized for this preprocessing.

Data pre-processing in NLP, akin to reducing outliers in traditional statistical methods, involves removing textual inconsistencies. However, it's crucial to acknowledge that this can lead to the loss of certain information. While it is true that some nuances might be lost in this process, our aim was to ensure the most effective and accurate analysis possible given the current state of the art.

Table 1
Methods applied for data pre-processing.

Removal/Fixation	Tasks
URLs	All URLs, regardless of the protocol (HTTP or HTTPS), were extracted and eliminated.
Mention and Hashtag	Mentions (e.g., '@username') and hashtags (e.g., '#trending') were identified and removed.
Email Address	Any detected email addresses were extracted.
HTML Tags	All HTML tags embedded in the text were stripped out.
Numerical Data	Pure numerical data were identified and removed.
Contraction	Common contractions like "don't" and "can't" were expanded to "do not" and "cannot" respectively.
Characters	Special characters, excluding apostrophes, were removed.
Case Normalization	The entire text corpus was converted to lowercase for consistency.
Stop Words	Common English words, which don't typically contain important significance, were filtered out.
Lemmatization	Words were transformed to their base or dictionary form to ensure consistency in representation.
Short Word	To further refine the data, words that have less than three characters were removed.
Space Normalization	All multiple white spaces were converted into a single space.

2.2.2. Text tokenization

Before introducing data to the model, it's essential to transform raw texts into a machine-readable format [36,51], a step termed as tokenization. In conjunction with the RoBERTa model, we deployed its associated tokenizer, skilled at converting input sequences into model-compatible tokens. Beyond fragmenting texts into words or sub-words, this tokenizer ensures sequences are equipped with the specific tokens mandated by the RoBERTa framework. Leveraging the RoBERTa tokenizer from the transformers library, we divided the preprocessed text into distinct tokens. The tokenizer occasionally segments words further into smaller sub-word components, aiding in managing unfamiliar words and enhancing model adaptability. For RoBERTa base models, token sequences are limited to a maximum of 512.

2.2.3. Text summarization

When working with transformer models, there's an inherent limitation on the number of tokens or 'pieces' of text they can process in a single batch, which is usually capped at 512 tokens for models like BERT and RoBERTa. However, in practice, we work with 510 tokens to allow for the addition of two special tokens, [CLS] or Classifier and [SEP] or Separator. These are integral to the architecture's functionality. The [CLS] token, typically positioned at the sequence's outset, is utilized in models like BERT where its representation aids in classification tasks after being processed through the transformer layers. Meanwhile, the [SEP] token serves as a marker to separate different segments within a sequence. In contexts, where understanding the relationship between two sentences is crucial, the format might appear as: [CLS] Sentence 1 [SEP] Sentence 2 [SEP]. To handle text that exceeds this 510-token limit, we incorporated the BART (Bidirectional and Auto-Regressive Transformers) model for summarization. Developed by Facebook AI [52] and optimized for text summarization with weights from facebook/bart-large-cnn, this method ensures that longer texts are shortened but still retain their core sentiment and message.

2.2.4. Generating co-occurrence matrix

The Co-occurrence Matrix is a tool that captures the frequency of two terms appearing together in a defined context, such as within the same document, sentence, or a specific number of adjacent words [53]. In this matrix, each cell reflects the joint occurrence count of the terms denoted by its respective row and column. Hence, both the rows and columns of the matrix correspond to individual terms. This matrix is invaluable for discerning term relationships, semantic analyses, and generating word embeddings [54]. To construct this matrix from our text tokens, we employed the *co_occurrence_matrix()* function. It operates using a sliding window mechanism, assessing the joint appearance frequency of two tokens within a predetermined range. By default, we adopted a window size of 5, examining five tokens both preceding and succeeding the current token in focus.

2.2.5. Sentiment analysis using transformer deep learning approach – RoBERTa

We employed the RoBERTa model, a variant of the BERT model, for sentiment analysis. Given the nature of our data, originating from X, it was crucial to employ a model that comprehends the nuances and semantics of tweets, which often differ from conventional textual data due to brevity, slang, and other platform-specific linguistic attributes. To this end, we chose the version of RoBERTa that was specifically pre-trained on a sentiment analysis task using the "cardiffnlp/twitter-roberta-base-sentiment" weights [55]. This specialized version of RoBERTa is adept at capturing the underlying sentiment of tweets, which can often be layered and subtle. Each processed text input was fed into the RoBERTa-based sentiment analysis pipeline. The model's output provided a sentiment label ('Negative', 'Neutral', or 'Positive') along with a confidence score. It's worth noting that any text exceeding the token threshold was first summarized before sentiment classification, ensuring it fit within the model's token limit. Ultimately, the dataset is

augmented with sentiment labels and their corresponding confidence scores as new columns.

2.2.6. Visualizations

A. Word Cloud Visualization

In our methodology, we employed the 'WordCloud' library in Python [56] for word cloud visualization, a prevalent technique in text analysis to underscore key terms associated with the main queries, herein, 'Stay-at-home,' 'HVAC,' and 'Energy Bill.' By depicting word frequency with varied font sizes and colors, word clouds facilitate the rapid identification and visual depiction of dominant themes within a text corpus. This method yields an intuitive overview of the prominent content, producing a graphic that is informative and comprehensible, thus rendering intricate data sets approachable to a broad audience spectrum, encompassing both specialists and laypersons.

B. Embedding visualization

To delve deeper into the linguistic patterns captured by the RoBERTa model, we visualized its embeddings. In NLP, embedding visualization represents the graphical depiction of word or sentence meanings in a vector space [54]. These embeddings are dense vector representations that encapsulate semantic significance within a multi-dimensional arena [57]. However, due to the sheer complexity of these vectors (e.g., 768 dimensions for a base RoBERTa model), direct visualization isn't practical. As a solution, we employed dimensionality reduction methods to represent these vectors in a more visually digestible 2D or 3D space. We employed the following steps for embedding visualization:

We utilized the base RoBERTa model, loading it with pre-trained weights ("cardiffnlp/twitter-roberta-base-sentiment"). To capture the semantic essence of input texts, we devised an 'extract_embeddings()' function, which, when given a list of processed texts, extracts their corresponding embeddings from the last hidden state of RoBERTa. To categorize these embeddings, we first identified the ideal number of clusters using the elbow method [58]. By employing K-means clustering [59] iteratively (from 1 to 9 clusters in this study), we assessed the inertia or within-cluster sum of squares [60]. By plotting the distortions against cluster numbers, we identified the 'elbow'—the point of inflection—indicating the optimal cluster count. Post identification, K-means [61] was used to group embeddings. To visualize these multidimensional embeddings, we applied t-SNE [62], a technique adept at condensing data dimensions while retaining inherent structures. We parameterized t-SNE for a 2D and 3D projection, fine-tuning perplexity as needed.

C. Attention score visualization (Attention Map) and interpretation

Attention visualization in transformer models, such as RoBERTa, is a technique that offers a graphical insight into how the model's attention mechanisms work when interpreting a given text [63]. This core component of transformer models assigns weightage to different text parts, allowing the model to prioritize some words over others based on the context. It is through this that relationships between words and their evolution through the model's layers can be observed. These visualizations play a pivotal role in understanding the model's behavior, highlighting any underlying issues, and providing avenues for potential performance enhancements. The study emphasized analyzing a text related to the significance of indoor air quality to human health during the pandemic, sourced from a random tweet. The attention scores indicate the emphasis placed on various tokens during processing, with high scores indicating strong token relationships. This begins with Self-Attention and gradually encompasses Neighboring Token Attention, Special Token Attention, and Distributed Attention. As we move through the transformer's layers, the initial layers spotlight basic syntactic

connections, while the higher layers unravel more complex, abstract relationships, forming a comprehension continuum from syntactic to semantic understanding.

RoBERTa's attention visualization is typically structured across multiple layers, usually 12, each possessing distinct attention scores. These layers, termed as 'leaders' here, are instrumental in grasping the model's hierarchical text processing and abstraction levels. As we move through these layers, the model focuses on diverse text aspects, constructing intricate representations. Lower layers may discern basic syntax, while advanced layers perceive more abstract semantics. These attention scores across layers can be visualized using heatmaps or directed graphs, illustrating attention strength and directionality among text words. Comparing attention patterns across these layers allows an understanding of the model's evolving text comprehension and which layers are vital for discerning certain relationships. Such visualizations are indispensable for deconstructing the transformer model's multi-tiered attention system, offering a lucid understanding of its text processing methods. For generating these attention visualizations, the Python package 'bertviz' was employed.

3. Results and discussion

Table 2 presents an overview of number of tweets based on sentiment categories across New York City boroughs for three distinct topics: 'Stay at Home', 'Energy Bill', and 'HVAC'. It highlights the sentiment distribution, including Positive, Negative, and Neutral sentiments, in Manhattan, Queens, Brooklyn, and the Bronx. Notably, the data illustrates that for the 'Stay at Home' and 'Energy Bill' topics, Manhattan has a substantial number of Positive sentiments, while Brooklyn and the Bronx exhibit higher Negative sentiments. On the other hand, for the 'HVAC' topic, the majority of tweets across all boroughs are Neutral, with Manhattan having the highest count of Positive sentiments. The limited data from Staten Island did not meet our threshold for a robust and representative sample. To maintain the integrity and reliability of our results, and to avoid potential bias that could arise from an insufficient data sample, we made a considered decision to exclude Staten Island from these specific analyses in our study.

The presented histograms in Fig. 2 detail sentiment analysis results related to the term "Stay-at-Home" for all boroughs. In the cumulative sentiment histogram, a symmetrical pattern emerges with the confidence score predominantly clustering around the 0.7 mark, indicating a balanced, moderate-to-high confidence for sentiments across the dataset. The histogram for negative sentiments reveals an intriguing pattern, exhibiting a skewed bimodal distribution with significant peaks at around 0.55 and 0.85, hinting at two prominent clusters of confidence levels for negative sentiments. The neutral sentiment distribution appears Gaussian, centralized around the 0.7–0.8 confidence interval,

Table 2
Number of tweets tagged with sentiment categories across NYC boroughs.

Tweet Count	Borough				Grand Total
	Manhattan	Queens	Brooklyn	Bronx	
'Stay-at-Home'					
Positive	225	79	106	21	431
Negative	217	94	180	65	556
Neutral	555	212	375	119	1,261
Total	997	385	661	205	2,248
'Energy Bill'					
Positive	97	7	12	2	118
Negative	47	6	17	8	78
Neutral	654	26	69	11	760
Total	798	39	98	21	9,56
'HVAC'					
Positive	142	19	32	5	198
Negative	79	7	19	6	111
Neutral	1,409	85	104	24	1622
Total	1,630	111	155	35	1,931

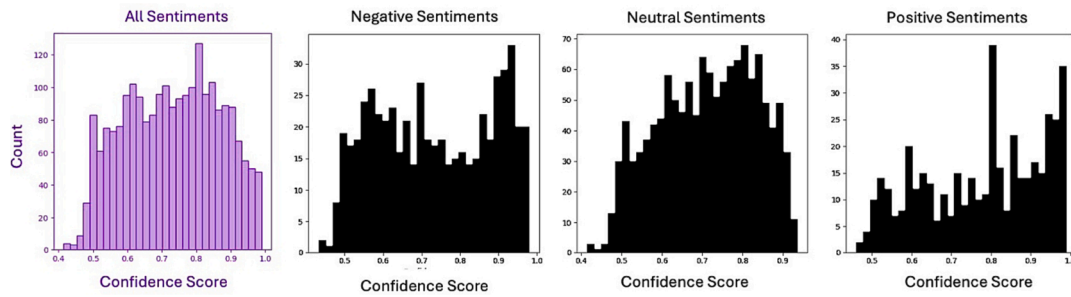


Fig. 2. Sentiment Analysis Histograms of 'Stay at Home' term, representation overall, negative, neutral, and positive sentiment confidence scores spanning major NYC boroughs.

suggesting a consistent and predictable pattern for neutral sentiments. Contrarily, the positive sentiment displays a more complex distribution, with distinct peaks at intervals such as 0.5, 0.65, and 0.9. This pattern highlights the heterogeneous nature of confidence in positive sentiment categorization.

In Table 3, sentiment analysis on tweets from the X Platform across various terms reveals a multifaceted public perception shaped by the prevailing societal context. Sentiments about 'Stay-at-home' range from positive views on the comfort of home (98.71 % confidence), to neutral acknowledgment of the importance of social distancing (63.07 %), and to negative feelings due to the challenges of NYC quarantine (50.81 %). For 'HVAC', sentiments are positively associated with enjoyment of improved outdoor gym ventilation (95.11 %), neutrally discussed in terms of the scientific aspects of energy and maintenance (83.35 %), and negatively viewed when there are frustrations with system inadequacies (82.17 %). Regarding 'Energy Bill', sentiments are positively aligned with the transformative potential of the COVID-19 relief bill for sustainable energy (65.07 %), neutrally questioning the market's response to hypothetical policy changes (85.69 %), and negatively reflecting shock at high charges (94.94 %). These tweets illustrate the diverse reactions to each topic over time and the varying degrees of confidence in the sentiment expressed.

Fig. 2 provides a representation of the inner workings of the

Table 3

Table 3. Representative samples of actual tweets for each term and sentiment category sourced from X platform.

Term	Tweet	Sentiment Category	Confidence Score
'Stay-at-home'	'happy see great relaxing time home enjoy stay safe'	Positive	98.71 %
	'exactly social distancing stay home vital fighting virus'	Neutral	63.07 %
	'nyc quarantine lost count day stay home single people'	Negative	50.81 %
'HVAC'	'fresh air good find gym windy tent back yard awning supercharged postcovid hvac great take class light weight metcon hiit'	Positive	95.11 %
	'science behind energy consumption built environment hygiene hvac system maintenance age covid'	Neutral	83.35 %
	'hope serious renovation hvac system non existent could catch f*** stroke'	Negative	82.17 %
'Energy Bill'	'covid relief bill massive investment clean energy tech mitigation strategy one wanted pandemic presenting chance transform economy sustainable drive growth long term save planet'	Positive	65.07 %
	'would mkt look like victory bond made energy bill'	Neutral	85.69 %
	'almost choked reading energy bill month f*** trying pull'	Negative	94.94 %

RoBERTa model, a variant of the BERT architecture, as it processes a short X post. The matrix of colorful graphics on the left highlights the model's multi-head attention mechanism across its layers. Each square in this matrix represents an "attention head," with the rows denoting different layers of the model, from 0 to 11, and the columns representing individual heads within each layer. A key feature of transformers such as RoBERTa is their ability to weigh the importance of different words in a sentence in relation to one another. The colors and patterns within each attention head depict the strength and direction of these attention weights. For instance, certain words may strongly attend to other specific words, leading to distinct color patterns. On the right side of the Fig. 3, we provided with detailed breakdowns of attention patterns for three specific layers: 1, 5, and 10. These breakdowns take the form of network diagrams, where words from the X post, such as "stay", "home", "guarantee", "breathe", "sick", and "air", are nodes interconnected by lines. The thickness and color intensity of these lines illustrate the strength of attention between the respective words. For example, in Layer 10, the word "home" appears to have a strong attention connection with "stay", showcasing the model's ability to recognize and emphasize the contextual relationship between these two words in the sentence, emphasizing the model's capacity to capture intricate semantic structures.

Fig. 4 represents neuron activations in a RoBERTa neural network model, given the token-style inputs. These activations illustrate how the model's neurons respond to specific input tokens related to the sentiments. The tokens, which include "stay," "home," "guarantee," "breathe," "sick," "air," and punctuation, suggest key thematic elements, with "<s>" and "</s>" denoting start and end tokens. The color spectrum indicates activation intensity, with lighter colors representing higher activation. Notably, the term "stay" exhibits high self-activation, underscoring its prominence. The word "home" shows a significant activation with the "guarantee" neuron, which may hint at the public's desire for safety assurances during the pandemic. Similarly, the connection between "breathe" and "sick" reflects the respiratory challenges of COVID-19. Other tokens, such as "air," have more uniform activations, suggesting a broader semantic significance. Collectively, the matrix provides an insight into RoBERTa's processing of pandemic-related terms, and its learned associations based on training data. This resonates with discussions on sentiments during the pandemic era, highlighting the intricate relationships and emotions experienced during this period.

Fig. 5 illustrates the clusters identified using the k-means clustering technique, derived from the RoBERTa embeddings of tweets pertaining to the "Stay-at-home" keyword. The original embeddings from the RoBERTa model represent each tweet in a 768-dimensional space (specific to the RoBERTa-base model). It's worth noting that there is a larger variant of RoBERTa, known as RoBERTa-large, which represents data in a 1,024-dimensional space, but in this analysis, the base version with 768 dimensions was utilized. The t-SNE technique was employed to reduce this high-dimensional data into more interpretable 2D and 3D spaces. On the left, the 2D representation depicts the distinct separation

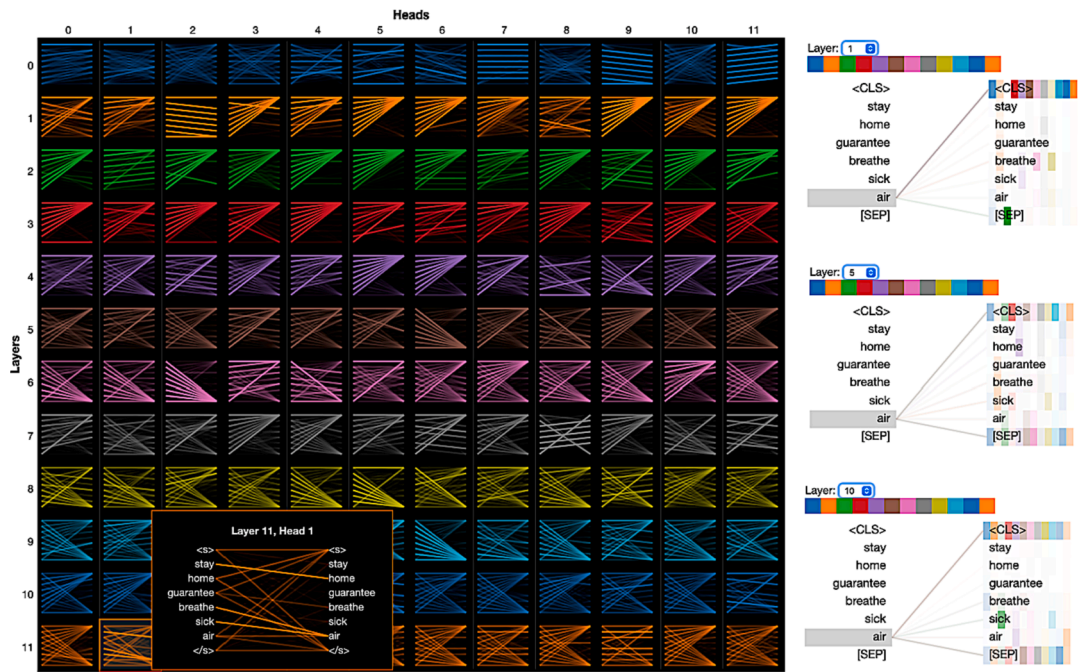


Fig. 3. Attention patterns of RoBERTa when processing a sample tweet, ‘stay home guarantee breathe sick air’, extracted from the X platform. On the left, the matrix showcases the attention scores across 12 layers (vertical) and 12 heads (horizontal), each color-coded based on the strength and direction of attention. On the right, detailed breakdowns of attention patterns for three specific layers of 1, 5, and 10 are illustrated.

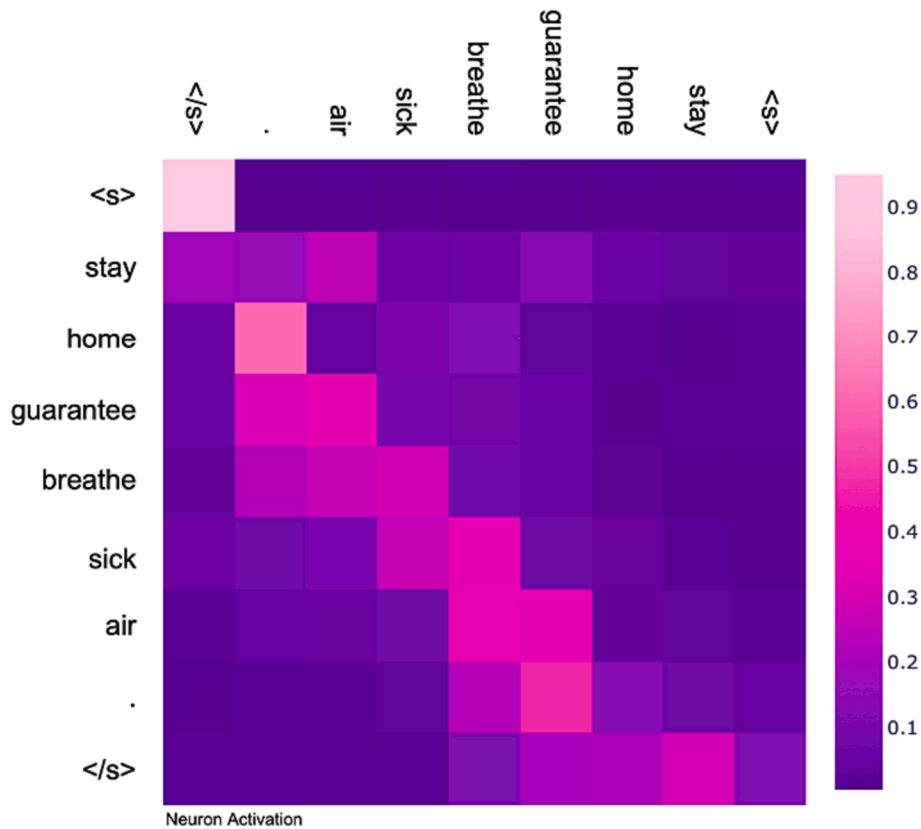


Fig. 4. Neuron Activation matrix of the RoBERTa model illustrating token-wise responses to the ‘stay-at-home’ term, revealing intricate learned associations between words during the COVID-19 era.

between various clusters, with some overlap hinting at the nuanced relationships between certain tweets. The 3D representation on the right offers a more intricate view of the data, allowing for the visualization of

distinctions between closely related clusters in the 3D space. Notably, certain clusters are more densely populated, suggesting dominant themes or patterns in the dataset, while sparser clusters might represent

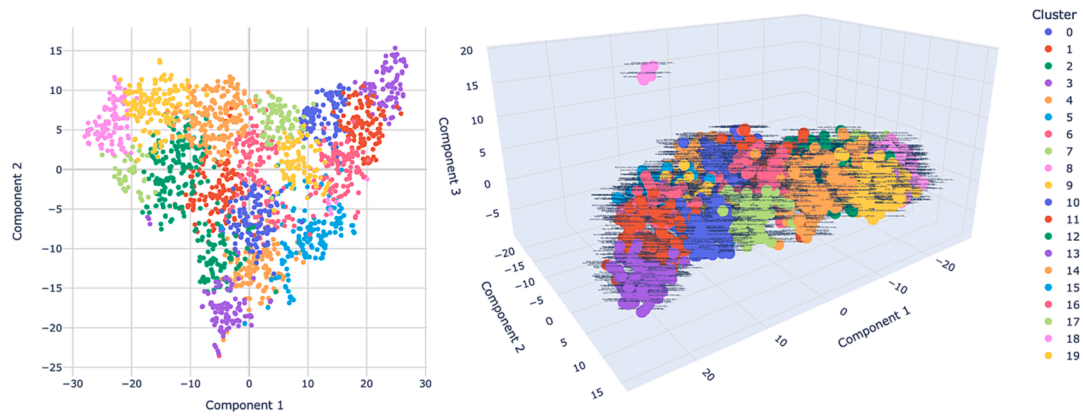


Fig. 5. T-sne visualizations of roberta embeddings from X data on the 'Stay-at-home' keyword, clustered using k-means method with 20 clusters (0–19): showcasing the diverse range of sentiments and topics in 2D (left) and 3D (right) data spaces.

more unique or outlier sentiments. With 20 distinct clusters, it becomes evident that there's a diverse variety of human perspectives, sentiments, and topics surrounding the "Stay-at-home" conversation on X platform. For instance, some clustered posts, such as those in cluster 7, reveal negative sentiments related to the impact of 'Stay-at-home' on job-related aspects. On the other hand, another cluster, like cluster 11, highlights discussions on the benefits of staying at home for health concerns. The use of RoBERTa embeddings, as opposed to traditional word cloud and network analysis, allows for a deeper understanding of the semantic relationships between tweets, capturing context and nuances that simpler methods might overlook [55].

Fig. 6 presents the temporal dynamics of sentiments linked to the term 'Stay-at-Home' across four New York City boroughs—Bronx, Brooklyn, Manhattan, and Queens—during the first year of the pandemic. Each sentiment—negative, neutral, and positive—is plotted monthly, offering a granular view of public mood swings. Across the board, April 2020 stands out as a pivotal moment for all boroughs, with a distinct spike in sentiment counts. This surge likely corresponds to the initial imposition of stay-at-home orders and the public's reaction to the sudden lifestyle shift. The uniformity of this spike across boroughs emphasizes the city-wide impact of the pandemic during this period, reflecting a heightened sense of awareness, adaptation, and possible anxieties associated with the term 'Stay-at-Home'. Following April 2020, there's a noticeable decline in sentiment counts, with periodic fluctuations. The persistence of all three sentiment categories throughout the period suggests mixed feelings among residents. While some people might have struggled with being confined at home due to psychological and mental issues [64–66] or economic and unemployment burdens [67] or domestic violence [68,69], others might have expressed gratitude for the safety measures or found a way to establish a routine within the limitations [70].

Periods marked by heightened negative sentiments might coincide with spikes in COVID-19 cases or other significant pandemic-related

events, including experiences related to lockdowns. Our wordcloud analysis, as shown in Fig. 7, indicates that the term 'home order' is mentioned more frequently within negative contexts than positive ones. This suggests that the people largely perceived the 'Stay-at-Home' mandate with caution or anxiety, potentially arising from uncertainties, lifestyle restrictions, or its broader impact on everyday life and the local economic landscape. A report by [71] uses Gallup's Life Evaluation Index to illustrate a significant drop in Americans' assessments of their lives during the early pandemic. This can be inferred that 'Stay-at-Home' mandates and the broader effects of the pandemic had significant implications on personal well-being. During this period, human sentiments expressed on social media frequently mirrored the personal life changes brought about by the pandemic. The Pew Research Center conducted an empirical study in March 2020, surveying 11,537 U.S. adults. This study documented the shifts in Americans' lives during the early stages of the pandemic, touching upon changes in social activities, belief systems, and variations across genders and age groups and beyond [72]. The findings offer insights into the profound impact and the adaptations people made in response to their altered realities.

Examining these sentiments from the perspective of economic burdens, especially regarding energy expenses, offers a telling narrative. According to report by National Bureau of Economic Research (NBER) as detailed in the 'Powering Work from Home' paper (NBER Working Paper 27937) [73], data reveals that in the U.S., when compared to the same period between 2016 and 2019 and adjusted for weather variations, the second quarter of 2020 experienced a 10 % increase in residential electricity use. Simultaneously, there was a 12 % reduction in commercial electricity consumption and a 14 % decline in industrial usage. The report further points out that, between April and July 2020, Americans paid an additional \$6 billion for home electricity, nearly balancing the decrease in commercial and industrial demand, which can be indicated as negative sentiments in part in the early lockdown. Notably, large U.S. cities and their metro areas including New York City

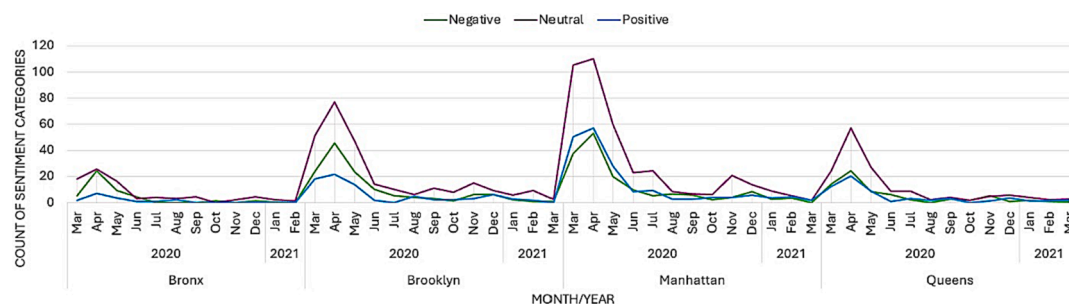


Fig. 6. Monthly sentiment counts (negative, neutral, positive) based on the term 'Stay-at-Home' segmented across various NYC boroughs (Bronx, Brooklyn, Manhattan, Queens) and years (2020, 2021) with a breakdown for each month, representing the first year of the pandemic.

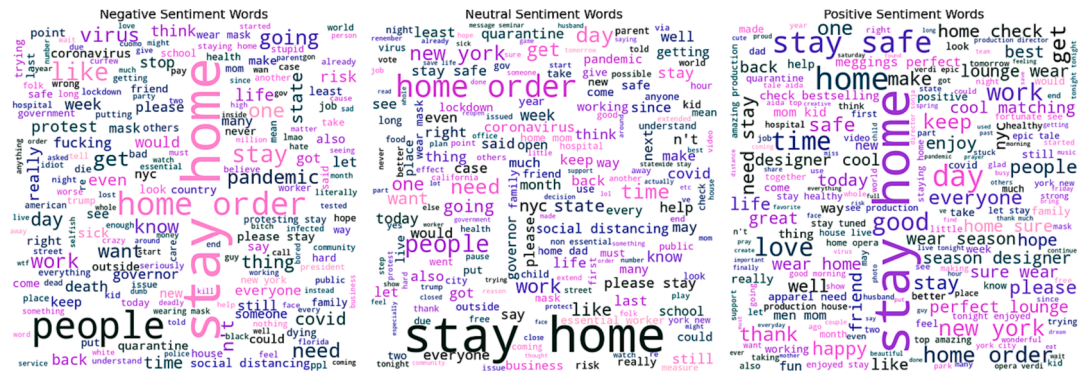


Fig. 7. A word-cloud visualization depicting sentiments (positive, neutral, and negative) associated with the term 'Stay-at-Home' for the first year following the initial lockdown. The frequency of the term 'home order' appearing in the negative-sentiment category, compared to the positive category, underscores the public's adverse response to the mandates imposed by authorities. (The size of the term on the graph indicates the frequency of its repetition in the posts).

metropolitan areas registered the highest anomalies in residential electricity consumption from April to July 2020 (compared with previous years), standing out from other locations.

From an intra-city perspective, Fig. 8 (a) depicts the normalized sentiment scores related to the "Stay-at-Home" term aggregated from March 2020 to March 2021, across different boroughs. Interestingly, tweets from Manhattan reflected the highest positive sentiment towards the term at 50.91 %, followed by Queens at 45.68 %. In contrast, the boroughs of Brooklyn and the Bronx exhibited a less favorable stance, recording positive sentiments at 37.04 % and a notably lower 24.42 %, respectively. Although Queens and Manhattan displayed higher positive sentiments, they still had significant negative scores of 54.32 % and 49.09 %, respectively. These insights reveal diverse reactions and emotions towards the 'Stay-at-Home' experience across the boroughs, providing a nuanced understanding of similar public sentiment during this period but with different magnitudes.

The negative sentiments about 'Stay-at-Home', especially in the Bronx borough, can be juxtaposed with the increased concerns about energy bills. For the term 'Energy Bill', we utilized the RoBERTa framework to perform sentiment analysis, applying the same methodology. The findings show a consistent spatial variation in sentiment across the boroughs, as illustrated in Fig. 8 (b) for the 'Stay-at-Home' term. Residents in Manhattan exhibited the most positive sentiment at 67.36 %, suggesting a largely positive perception or experience related to energy bills. Historically, the Bronx has had areas with greater poverty levels than other boroughs in the city [74]. In light of these findings, a significant drop in positive sentiment (by 20 %) towards energy bills in the Bronx may indicate an intensified financial burden faced by its residents in managing utility costs during that period.

Fig. 9 offers an in-depth view of sentiments on pertaining to 'Energy Bills' across the major boroughs of New York City for eight consecutive years from 2015 to 2022, emphasizing the significant impact of the pandemic in the year 2020. Notably, while varying sentiments are exhibited across the boroughs, The Bronx, which has the lowest average

income, stands out with a striking 100 % negative sentiment in 2020, reflecting the profound dissatisfaction of its residents concerning energy bills during the intensive stay-at-home phase of the pandemic. This period saw higher occupancy rates in residential buildings due to the order and, in many cases, no employer contributions to energy bills for work-from-home modality. As the study progressed into 2021 and 2022, the results illustrate a shift towards more neutral and balanced sentiments. This evolution aligns with the gradual resumption of normal life and potentially lower energy expenditures as public spaces reopened and people returned to their workplaces.

Manhattan, which has the highest average income, largely hovers within neutral sentiments, but a slight increase in negative sentiments is present in 2020. Queens and Brooklyn show reduced negative sentiments toward 'Energy Bill.' This anomaly could be attributed to reduced energy consumption in areas like transportation, as residents in these boroughs typically travel longer distances to reach City's Central Business District (CBD) which is located in downtown Manhattan. This may have seen a decline in negative sentiments due to increased remote work practices during that year. Additional research would be needed to verify this hypothesis. The connection between urban building and transport energy use patterns and CBD has been thoroughly demonstrated in a comprehensive framework by [75] focusing on Chicago, IL. The absence of negative sentiment in 2016 for 'Energy Bill' discussions in Manhattan may reflect a period-specific trend, possibly could be influenced by a variety of factors, such as beneficial policy changes, favorable economic conditions, or a shift in public interest to other pressing issues during that year. Further investigation is needed to fully understand this anomaly within the context of our dataset. Overall, this finding accentuates the probable financial and emotional challenges New Yorkers faced, especially in The Bronx, where energy bills became a focal point of distress during the pandemic.

Fig. 10 corroborates these findings through data extracted from the DOE's LEAD Tool. The Bronx experiences the highest energy burden at 4, indicating that, on average, households in the Bronx allocate a more

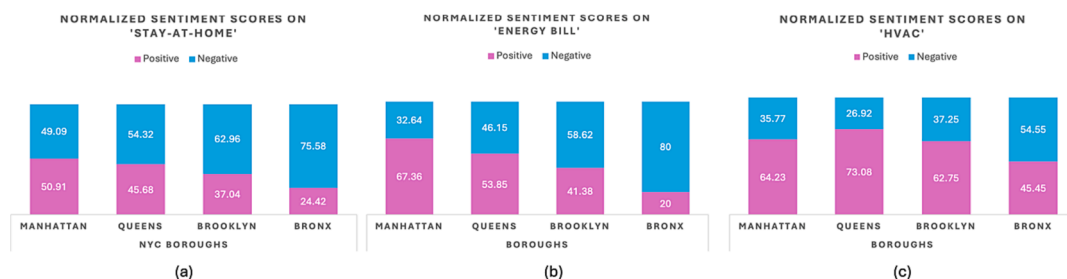


Fig. 8. Normalized sentiment scores for 'Stay-at-Home' (a), 'Energy Bill' (b), and 'HVAC' (c) across various New York City boroughs. Data was extracted from the X platform and sentiment analysis was conducted using the RoBERTa algorithm.

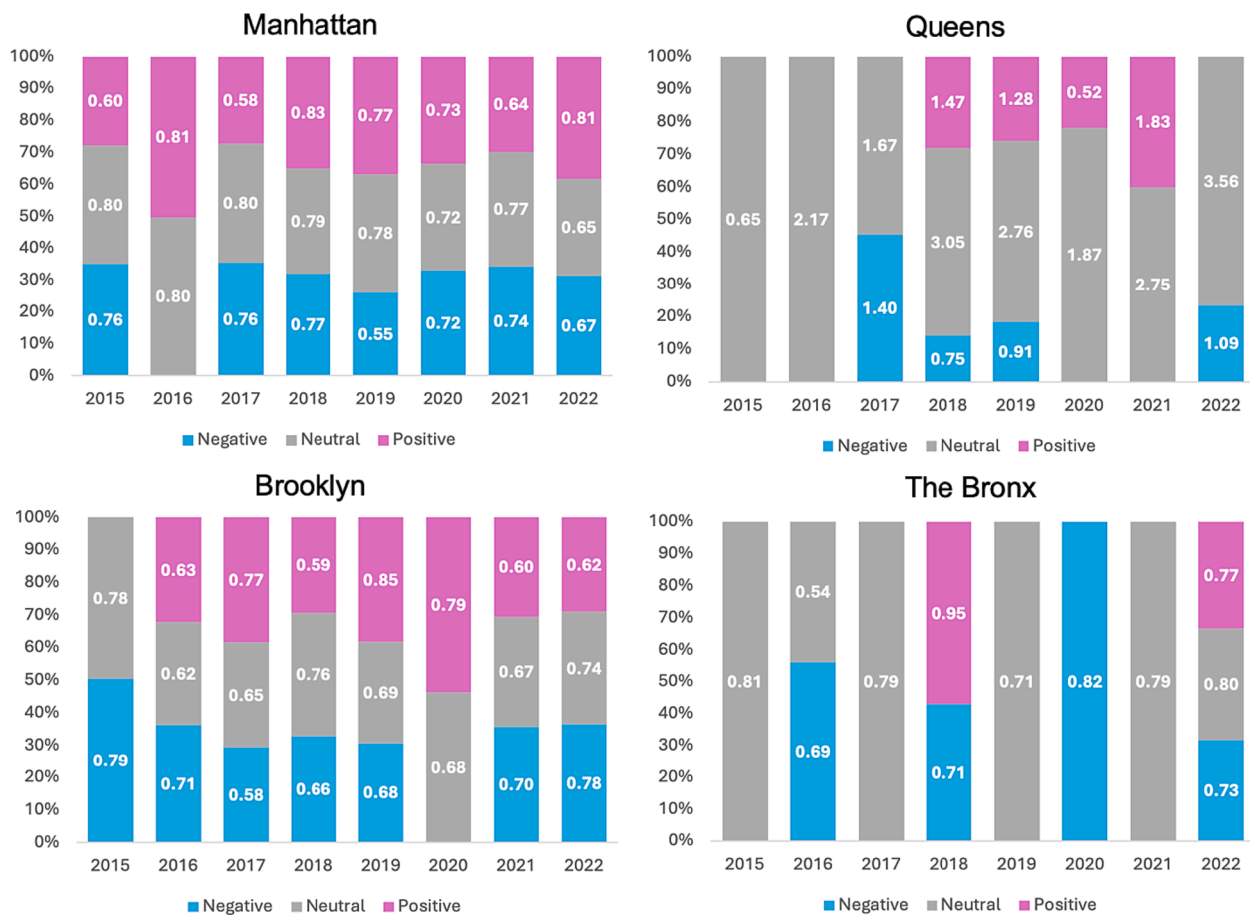


Fig. 9. Time series averaged sentiment scores –with normalized values between 0% and 100%– of ‘Energy Bill’ tweets across NYC boroughs (2015–2022), highlighting the changes in negative sentiments during the 2020 stay-at-home era.

significant portion of their income to energy expenses. In contrast, Manhattan has the lowest energy burden at 1, suggesting that its residents generally commit a smaller percentage of their earnings to cover energy costs. This visualization further offers a comprehensive overview of New York City’s residential energy consumption patterns, differentiating by borough, the relative energy burden on households, the vintage of buildings, and the primary heating fuel type used. Notably, there’s a visible trend where older buildings, particularly those constructed before 1940, which are predominant in boroughs like The Bronx, have a higher energy burden, and they predominantly rely on less efficient fuel types such as fuel oil and bottled gas. This suggests a link between building age, energy inefficiency, and the increased financial strain on residents, which is consistent with the sentiment analysis indicating a spike in negative sentiment during 2020, especially in The Bronx.

Further scrutiny of the urban building characteristics presented in Fig. 11 illuminates distinctions among the boroughs. The Bronx displays a concerning trend, with as many as 94.66 % of its single and two-family residences not having undergone improvements. This starkly contrasts with Manhattan, where only 54.42 % of such residences remain unimproved. This could imply that residents in the Bronx predominantly reside in potentially less efficient housing stock, which has not seen significant renovations or upgrades. When we assess the residential area relative to the building’s gross floor area, the Bronx, again, leads with 82.01 %, indicative of its residential-centric layout. In terms of mixed-use buildings, the Bronx has a higher residential space proportion compared to Manhattan, suggesting mixed-use buildings in the Bronx are more residentially dense. As for building types including single and two family, and apartments with and without elevator, while all

boroughs exhibit a predominant residential orientation, the Bronx edges ahead, reinforcing its vast residential character. The building characteristics in the Bronx, combined with the sentiments around energy bills, accentuate the intricate relationship between the nature of housing stock, its efficiency, and resident satisfaction. In essence, the data seems to suggest that the outdated or unimproved housing structures in the Bronx may be a significant contributor to the financial and energy challenges faced by its residents during the stay-at-home period.

Buildings’ energy consumption patterns, notably during peak periods, plays a pivotal role in determining the overall energy bill [76]. The stay-at-home directives during the pandemic further increased these consumption patterns, with HVAC systems being a dominant factor in escalating energy use in residential settings [77]. However, the ability to navigate these amplified energy demands does not impact all households equally. Socio-economic factors come into play, as research underscores that high-income households are more equipped financially to manage the burdens associated with heightened energy consumption [78]. Turning to the results of sentiment scores on ‘HVAC’ tweets (Fig. 8 (c)), Queens stands out with the highest positive sentiment of 73.08 %, suggesting a largely positive reception or experience with HVAC systems or services. Manhattan and Brooklyn follow closely with scores of 64.23 % and 62.75 % respectively, indicating general satisfaction. The Bronx demonstrates a considerably less positive sentiment (45.45 %). During the pandemic’s stay-at-home orders, the impacts of continuous occupation on indoor environmental quality (IEQ) and energy consumption became more pronounced, highlighting the significance of comfort and the potential implications for energy bills [79]. A well-functioning HVAC system becomes pivotal in ensuring such comfort. Notably, boroughs such as Manhattan and Queens, which exhibited more positive

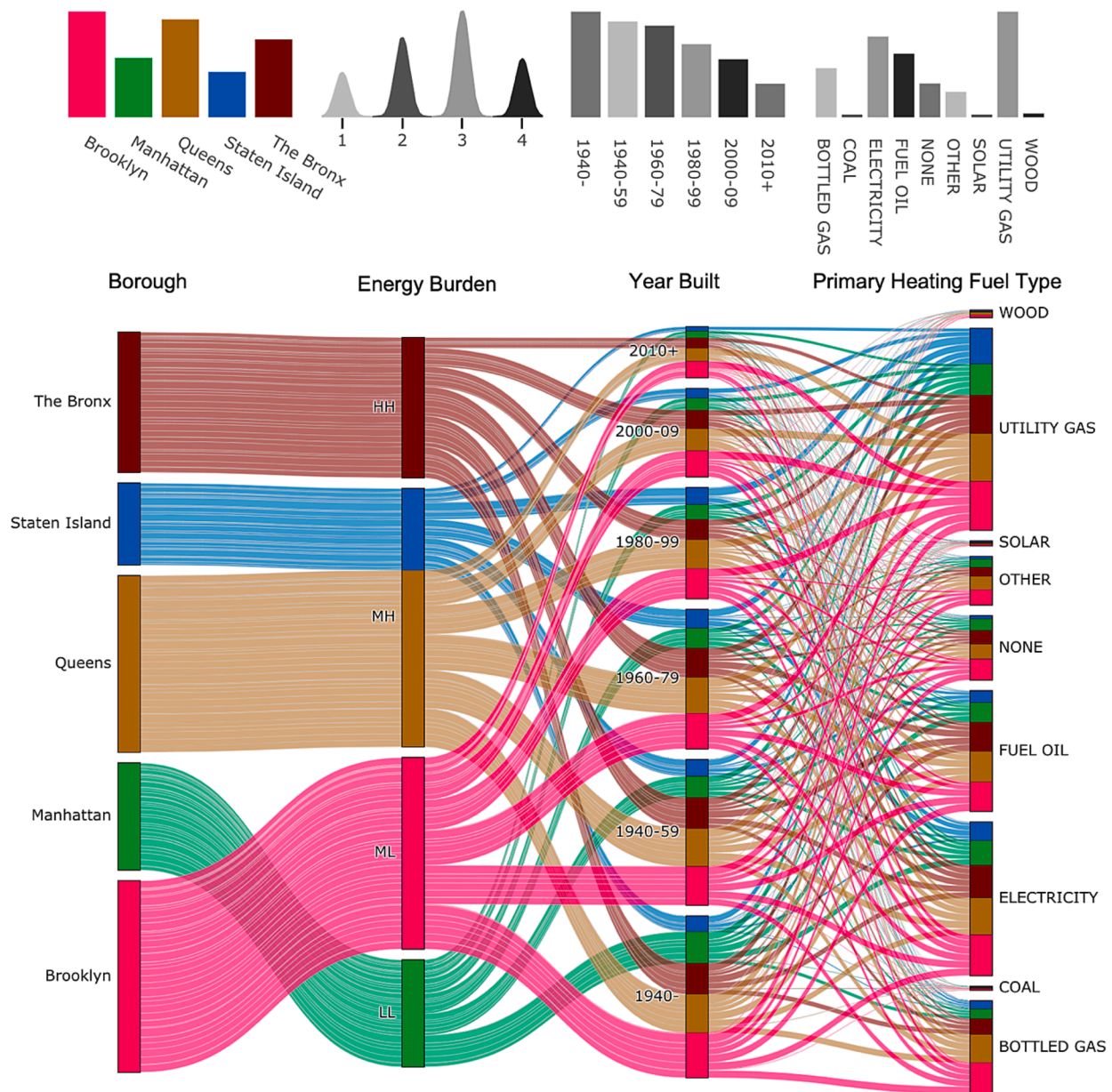


Fig. 10. Sneaky plot highlighting a visual correlation between energy burden, building age, and heating fuel type across NYC boroughs. Energy Burden is categorized as LL (Low-Low), ML (Medium-Low), MH (Medium-High), and HH (High-High) based on the income percentage spent on energy. The numbers 1 to 4 at the top represent these categories in ascending order of burden. Data sourced from the LEAD Tool.

sentiments towards the 'Stay-at-Home' term, also demonstrated relatively positive sentiments towards 'Energy Bill' and 'HVAC'. This suggests a possible link between financial stability, home comfort—potentially due to the affordability or efficiency of HVAC operational costs—and overall well-being during the lockdown. The sentiments during stay-at-home mandates across the boroughs further accentuate the spatial justice concerns. Areas with more negative sentiments towards the 'Stay at Home' order, might have faced compounded challenges, both in terms of economic strain from rising energy bills and potential health risks from inadequate or inefficient HVAC systems. It's not just about staying at home, but the quality of that stay, influenced by access to resources and the overall living environment.

Fig. 12 depicts the distribution of sentiments, both negative and positive, segmented by gender across four New York City boroughs: Manhattan, Queens, Brooklyn, and the Bronx. The sentiments are normalized as percentages, ensuring that the combined total of all sentiments for each borough equates to 100 %. The figure highlights a

unique trend in Manhattan, where positive sentiments among females surpass the negative ones. In contrast, in Queens, Brooklyn, and the Bronx, there is a significant tilt towards negative sentiments among females. This trend reaches its peak the Bronx, where the ratio of negative to positive sentiment among females spikes to 33.3 %, indicating the highest level of female dissatisfaction. This might be because of increase of work-related aspects among women [80] or violence against females reported during the COVID stay-at-home [81,82].

The escalation in negative sentiment among women, especially noticeable in the Bronx, is further highlighted by data reported in a Bronx community newspaper [83]. This data, sourced from police and district attorney offices, indicates that the Bronx leads all boroughs both in the total number of chronic cases of gender-based domestic violence—amounting to 797 in 2021—and in the rate of such cases per resident, underlining the gravity of this issue in the borough. This disparity between genders' sentiments may also be influenced by broader socioeconomic factors highlighted by [84] working paper. They

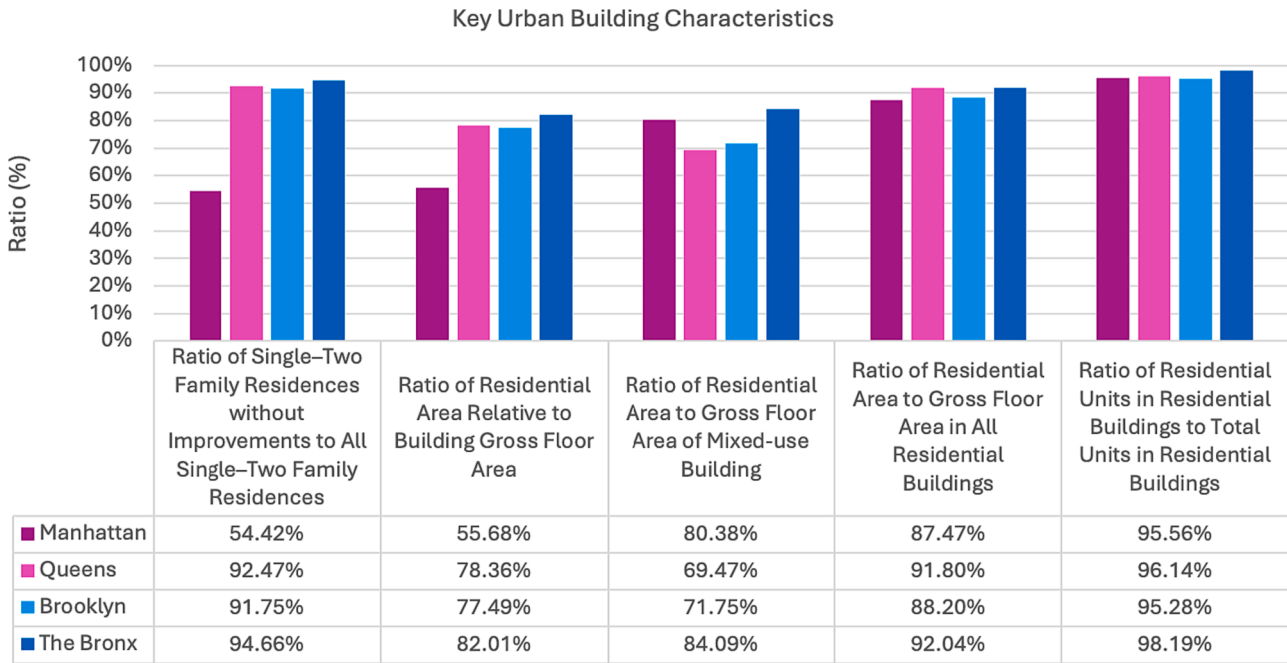


Fig. 11. Comparison of key urban building characteristics across NYC boroughs retrieved from the PLUTO (23.v2) dataset.

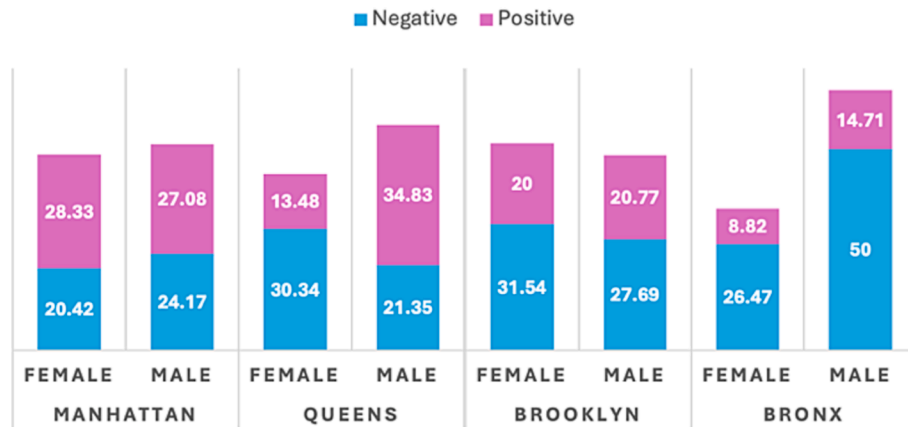


Fig. 12. Normalized sentiments score (%) based on gender across four NYC Boroughs.

point out that the COVID-19 pandemic's economic impact has had profound implications for gender equality. The downturn disproportionately affected sectors with high female employment, exacerbating job losses among women. Additionally, the closure of schools and day-care centers placed extra childcare burdens on working mothers, potentially having long-term effects on their career trajectories due to the high returns to experience in the labor market. These factors might contribute to the heightened dissatisfaction and negative sentiments among women, as reflected in the Bronx's gender-based violence statistics.

The study by [47] on overcooling in buildings and its gendered impacts on thermal comfort provides insights into how women's experiences during the COVID-19 stay-at-home periods might differ from men's due to varying thermal sensitivities. This research suggests that gender disparities in the built environment, initially observed in office settings, could extend to home settings. Furthermore, a review by [85] reinforces this perspective, revealing that women generally experience more dissatisfaction than men in cooler environments. Such differences in thermal comfort highlight a broader context of gender-specific experiences in both public and private spaces, indicating that the comfort

levels in residential buildings during the pandemic may have more significantly affected women, potentially influencing their views on stay-at-home measures and their overall living satisfaction.

4. Limitation and future research

In investigating sentiment across New York City boroughs using social media data, we must acknowledge certain constraints. The brevity and informal nature of social media posts, especially on platforms with character limits like X, may not fully encapsulate the diverse sentiments of a broader demographic. Additionally, social media users represent a specific subset of the population, which may not reflect the rich tapestry of New York City's socio-economic and cultural diversity.

Our study utilized the RoBERTa model for sentiment analysis but was limited to a sample of the data. Future research could benefit from a more robust validation mechanism, perhaps involving independent evaluation to improve sentiment categorization. The issue of echo chambers and bots on social media further complicates sentiment analysis, as they can distort the perception of widespread public opinion. While we attempted to filter out bot-generated content, the

possibility of their complete elimination is challenging, pointing to the need for advanced detection techniques.

Lastly, the study's focus was restricted to English language tweets and a single platform, potentially overlooking the sentiments expressed in other languages and on varied social media platforms such as Facebook, Instagram, and Reddit. Additionally, the study's scope was limited to residential energy concerns, not accounting for the transport sector's role in energy justice. Expanding the scope of data sources and sectors in future research could offer a more rounded understanding of the sentiments related to urban infrastructure and energy justice. Integrating this with multimodal data sources could pave the way for more informed urban policy and planning that is attuned to the nuanced perspectives of city residents.

5. Conclusion

In conclusion, it is evident how occupants' feedback posted on social media, building characteristics, and the dynamic socio-spatial fabric of New York City boroughs are intertwined. By delving into sentiment analysis, particularly around pivotal terms like "Stay at Home", "Energy Bill", and "HVAC", we've been able to unravel not just the immediate reactions of inhabitants, but also the broader repercussions in the context of Spatial Justice and, more pointedly, Energy Justice. The sentiment analysis, segmented by gender, adds another layer to the study, revealing distinct perceptions and experiences among different demographic groups. When juxtaposed with the PLUTO and LEAD databases, sentiments highlight stark inequalities in urban built structures, amplifying the disparities in infrastructure and access to resources. Such topics as Energy Justice spotlight societal imbalances, suggesting that particular boroughs and demographics disproportionately shoulder the consequences of energy inefficiencies and encounter barriers to sustainable energy alternatives. By closely scrutinizing the urban landscape and infrastructure, it's evident that these disparities pose genuine challenges for the inhabitants, influencing their well-being and their perceptions of their living environments. These inequalities also suggest systemic issues in urban planning and resource allocation, intrinsically linked to the overarching theme of Spatial Justice. By harnessing social media data and deploying NLP computations—further enhanced by the sophistication of transformer deep learning methodologies like the RoBERTa model—this study has not only captured the emotional nuances of the city but has also emphasized the pressing need for fair urban growth. This intricate exploration calls for urban planners, policy-makers, and researchers to address these crucial challenges, championing a vision for a more equitable and inclusive urban landscape.

Our findings can directly inform data-driven policy development by providing real-time insights into public opinion on energy-related issues as expressed on social media. While this data may not capture the entire spectrum of societal views, it offers significant insights for policy-making. These insights can guide the creation of targeted energy subsidies for communities burdened by high energy costs and influence policies that promote energy-efficient HVAC systems in new constructions and renovations. The sentiments expressed regarding 'Stay-at-Home' measures are particularly valuable for energy planning, highlighting the need for adaptable policies in response to shifts in domestic energy consumption patterns, as observed during the pandemic. Understanding the varied sentiments across boroughs is crucial for formulating equitable policies during the energy transition, ensuring that the impact on vulnerable communities is carefully considered. Additionally, our research can inform the development of energy education programs that focus on aspects of energy use and efficiency highlighted by our study. Finally, our methodology serves as a tool for ongoing monitoring of public sentiment, providing a basis for the continuous evaluation and refinement of energy policies to better align with community needs and contribute to energy justice.

CRedit authorship contribution statement

Mehdi Ashayeri: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Narjes Abbasabadi:** Conceptualization, Validation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors, Mehdi Ashayeri and Narjes Abbasabadi, declare that they have no financial or personal conflicts of interest related to their respective affiliations at Southern Illinois University at Carbondale and University of Washington at Seattle.

Data availability

The authors do not have permission to share data.

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